ECC3479 Final Report

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Our research question is: *How do changes in key macroeconomic variables affect stock returns of ASX-200 constituents from 2014-2024, and to what extent do these effects challenge the assumptions of the Efficient Market Hypothesis?* This research question is of high relevance as it examines whether stock prices in the Australian share market, immediately and completely, reflect macroeconomic information, which is a key assumption of the Efficient Market Hypothesis. By definition, the Efficient Market Hypothesis (EMH) is a financial theory that suggests that asset prices, like stocks, completely and immediately reflect all available information at any point in time. As prices immediately adjust to new information, this theory implies that consistently staying ahead of the market on a risk-adjusted basis is impossible (Fama, 1970). Analysing the ASX-200 businesses between the period of 2014-2024 highlights how central macroeconomic factors, like interest rates, exchange rates, inflation, GDP growth and unemployment, all impact stock returns over time.

Firstly, there have been many economic disruptors that have caused substantial market volatility between 2014-2024, most notably the COVID-19 pandemic, aggressive interest-rate hikes by the RBA, the war in Ukraine and Russia, global supply-chain disruptions and high inflation. These occurrences provide a strong real-world context for testing if Australian financial markets actually react immediately and behave efficiently.

Secondly, there is a practical importance of the accuracy of the EMH for investors, firms and policymakers. If macroeconomic variables consistently predict stock returns, we can assume that by using publicly available economic information, investors could gain abnormal returns, thus challenging the EMH. The real-world consequences of this could impact portfolio management, financial regulation and general investment strategies (Beechey et al., 2000).

Lastly, there is minimal research on the Australian market efficiency, as most financial literature and studies are based on the United States and European markets. This research may help us consider whether a smaller, relatively resource-dependent economy reacts differently to macroeconomic shocks in comparison with larger, more dominant global markets (Chiah et al., 2018).

The results for our analysis provided mixed evidence for both the semi-strong and weak-form of the Efficient Market Hypothesis (EMH), implying that (1) market information is included in stock prices to some extent however, it may not be immediately or fully incorporated into the price, and (2) past information is correlated with future information. Across the sample of ASX-200 stocks we reviewed, the impact of macroeconomic factors varied largely due to different industries, market sector exposures and firm types like value or growth stocks. The 4 main stocks we review are the *Commonwealth Bank of Australia* (CBA), *CSL limited* (CSL) which is part of the healthcare sector, the *BHP group* (BHP) in the mining and materials sector and finally the *Wesfarmers Limited (*WES) in the consumer discretionary or industrials sector.

Data

The data used for this analysis was gathered using R software packages, including Quantitative Financial Modelling Framework (quantmod) (Ryan & Ulrich, 2025), and Download and Tidy Data from the Reserve Bank of Australia (readrba) (Cowgill, 2025). Quantmod was used to download the financial data from Yahoo Finance simultaneously for all four ASX listed stocks, including the market benchmark ticketed ("AXJO") over the 10 year period from 2014-2024. Data was adjusted for daily analysis to satisfy the assumptions of EMH, and to create a better functioning time series sample. This data was then merged into one daily price series for consistency and modelling purposes. Log returns were computed, and they are essential to conduct many statistical tests including the Jarque-Bera, and the creation of behavioural variables. These were merged into the "reg_data" which was where all of the clean data was compiled for the Multi-Factor CAPM model.

Readrba was used to gather all of the data from the Reserve Bank of Australia (RBA) website. The series IDs used for this analysis were "FIRMMBAB90" (the 90 day bank bill rate for the RBA) and "FIRMMCRTD" (interest rate data). First the data was downloaded using these series IDs, then cleaned into the 2014-2024 period, and lastly created into a binary variable and merged into reg_data. The 90-day bank bill rate dataset was used for the systematic risk component of the CAPM model, a key component that must be satisfied.

Figure 1: Exemption Table

Step	Observations	Excluded
1. Raw Yahoo Finance Data	2785	0
2. Log-returns (omitted N/A)	2784	1
3. Matching date with ASX200 index	2780	4
4. Merging with RBA and Rf	2780	0

5. Final regression sample (dropped lags/momentum N/A)	2720	60

Figure 1 illustrates the change in number of observations across the data cleaning process. 65 observations in total were omitted, due to N/As and the structure of the behavioural variables (E.g. designing the 20-day lag period).

Figure 2: Summary Statistics of Log Returns for Securities and Index

Security	CBA	BHP	WES	CSL	ASX200
Minimum	-0.10543	-0.15565	-0.10431	-0.10929	-0.10203
1st Quartile	-0.00592	-0.00948	-0.00571	-0.00647	-0.00433
Median	0.00081	0.00033	0.00092	0.00069	0.00072
Mean	0.00045	0.00035	0.00053	0.00056	0.00015
3rd Quartile	0.00753	0.01077	0.00727	0.00812	0.00518
Maximum	0.12453	0.11283	0.10694	0.11769	0.06766
Std. Deviation	0.01309	0.01776	0.01260	0.01412	0.00948

Figure 2 demonstrates the summary statistics for the securities and the ASX200 index. On average, WES appears to be the highest performer on average with mean returns (0.00092) exceeding the market index and other stocks. The ASX200 index has the lowest standard deviation, this is expected due to the large number of stocks it contains. Contrary to the index, BHP has the highest standard deviation (0.01776) reflecting a high risk profile, with CBA exhibiting the lowest (0.01309). The difference between the third and first quartiles indicates that the data could be rightward skewed; this was formally confirmed by the Exploratory Data Analysis.

Empirical Strategy

Our analysis is descriptive and we only claim evidence of the existence of correlations and not casualty. We estimate a Multi-Factor Capital Asset Pricing Model (CAPM) to examine the impact of RBA monetary policy decisions (Rise, Cut, No Change) on stock returns. If the market is efficient, news of the RBA decision is considered to be ‘priced in’ meaning that we should not expect significant deviations from returns. Additionally, we control for the behavioural variables of momentum and overreaction, following research by (DeBondt and Thaler, 1985) and (Jegadeesh & Titman, 1993). Assuming the market is efficient, the previous day’s return should not exceed ±2 standard deviations from the mean (overreaction) and past stock performance over short horizons (20 days) should not predict future returns (momentum).

The original CAPM model is given by:

$$E[r_{ft}] - r_f = \alpha + \beta_j(E[r_m] - r_f) + \varepsilon_t$$

The original CAPM uses systematic risk as a main predictor for abnormal returns, given by market return minus the risk free rate. Under this specification, alpha should be equal to zero if the market is efficient, meaning that no security should earn higher returns beyond its given level of systematic risk. We expect Beta to remain significant throughout the analysis, as it measures the sensitivity of a security to adverse market conditions. The single-factor model has its limitations however, and additional explanatory factors such as the use of RBA dummy variables and behavioural observations (non-rational risk pricing) extends the original CAPM in our context over a 10-year period.

Multi-Factor CAPM model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \gamma_1 Hike_t + \gamma_2 Cut_t + \delta_1 Hike\_Post1_t + \delta_2 Cut\_Post1_t + \lambda_i Mom_{i,t} + \phi_i LargePos_{i,t} + \theta_i LargeNeg_{i,t} + \mathcal{E}_{i,t}$$

Where:

- $R_{i,t}$ is the return on an individual asset at time t.
- $R_{f,t}$ is the return on a riskless asset at time t.
- α (Alpha) refers to an asset’s ability to earn abnormal returns relative to the market return. If alpha is not equal to zero, this would imply that an asset is mispriced which contradicts efficient market hypothesis.
- β (Beta) is the coefficient measuring the sensitivity of the security to market movements. It is not relevant as this is a risk-return profile.
- $(R_{m,t} - R_{f,t})$ is the market return minus the risk-free rate. This helps explain the additional compensation investors receive for taking on risk, and is a fundamental component of the CAPM model.
- $Hike_t$ is a binary variable equal to 1 on the days during the period that the RBA raised the interest rate, 0 if opposite.
- Cut_t is a binary variable equal to 1 on the days during the period that the RBA cut the interest rate, 0 if opposite.
- $Hike\_Post1_t$ is a binary variable equal to 1 on the trading day after the RBA raised the interest rate to account for a lag in reaction to information, 0 if opposite.

- Cut_Post1_t is a binary variable equal to 1 on the trading day after the RBA cut the interest rate to account for a lag in reaction to information, 0 if opposite.
- $Mom_{i,t}$ is the 20-day cumulative log return for a given stock, lagged by a trading day. It is the first behavioural variable in this model documenting momentum bias (Jegadeesh & Titman, 1993).
- $LargePos_{i,t}$ is a binary variable equal to 1 if yesterday's return exceeded +2 standard deviations from the mean, 0 if opposite. This documents the overreaction bias (De Bondt & Thaler, 1985).
- $LargeNeg_{i,t}$ is a binary variable equal to 1 if yesterday's return was below -2 standard deviations from the mean, 0 if opposite. This documents the overreaction bias (De Bondt & Thaler, 1985).
- $\varepsilon_{i,t}$ (Error term) is the unexplained portion of the stocks return on day t. It is commonly used as a measure for idiosyncratic risk, but it will be interpreted as omitted factors.

The extended CAPM model allows for a simultaneous test of weak form EMH through the behavioural variables, and a semi-strong form through the absorption of macroeconomic information from the RBA. The analysis will use a 5% level of significance.

Regression results

Significant Factor / Variable Name	Coefficient Estimate (Std. Error)
BHP	
Market Excess Return (β)	1.21902*** (0.02736)
Rate Cut	-0.00929* (0.00453)
Momentum	-0.00851* (0.00358)
CBA	
Market Excess Return (β)	1.10940*** (0.01591)
Post Hike Day 1	-0.00435* (0.00219)
Overreaction (+)	0.00257* (0.00113)
CSL	
Market Excess Return (β)	0.83820*** (0.02398)
Overreaction (+)	-0.00424** (0.00160)
Overreaction (-)	0.00590*** (0.00150)
WES	
Alpha (Intercept)	0.00046* (0.00019)
Market Excess Return (β)	0.89512*** (0.01893)

Values display point estimates followed by standard errors in parentheses.
R-squared: CBA (0.6431), WES (0.4523), BHP (0.4245), CSL (0.3203)
*** p<0.001, ** p<0.01, * p<0.05 based on standard two-tailed OLS t-tests.

Figure 3: Regression results

Our CAPM regression estimated significance for the alpha, beta, RBA, and behavioural variables. Our findings show heterogeneity, and will be explored in more detail.

α coefficient: The Multi-Factor CAPM returned one significant alpha for WES, the other stocks did not show significance for this coefficient. From figure 3, WES is expected to have a return that is 0.0464% per day higher than the CAPM predicts. A positive alpha indicates that the stock returns exceeded CAPM expectations during this period, therefore WES outperforms its expected return at the level of systematic risk it faces. Under CAPM and semi-strong market efficiency, alpha is expected to be statistically indistinguishable from zero, thus implying that this finding is stock-specific.

β coefficient: As anticipated, all stocks returned significant beta coefficients. Figure x illustrates the different risk properties of each security to market-wide movements. When $\beta > 0$, a stock is more sensitive to systematic market risk, while $\beta < 0$ suggests lower exposure relative to the broader market. We find **that** a 1% increase in market excess returns is associated with an estimated 1.22% increase in BHP excess returns and a 1.115 increase in CBA excess returns, ceteris paribus. In contrast, WES and CSL exhibit beta coefficients below one, implying comparatively lower sensitivity to market wide fluctuations. These differences can be attributed to the varying industries and their associated risk profiles. For example, BHP relies on a stable exchange rate due to its exposure to commodity market and global economic conditions, and therefore is susceptible to unfavourable macroeconomic movements relative to other firms not in the mining industry.

RBA coefficient: CBA and BHP coefficients are significant at the 5% level. No other stocks exhibited significant RBA coefficients, implying that information from the dummy variables was priced efficiently. For the CBA, ceteris paribus its excess returns are on average 0.43% lower on the day following an RBA rate increase. The lagged significance shows that information was absorbed a day after the hike occurred, illustrating a violation of semi-strong form EMH. On the other hand, BHPs excess returns are on average 0.93% lower on days when the RBA has cut the interest rate. Since this variable is not lagged, it supports EMH as information has been reflected on the day.

Behavioural coefficients: CSL, BHP and CBA are the only significant behavioural variables. Overreaction bias was observed for CSL, with an inverse relationship suggesting that after a large positive or large negative day, returns will move in the opposite direction. This supports weak-form EMH because past prices (the extreme days) can predict future returns, thus not reflecting current information. On the other hand, BHP returns follow a cycle of regressing towards the mean. Similar to the statistical concept, a one unit increase in the 20-day cumulative log return is, ceteris paribus, associated with a 0.85 percent decrease in BHP excess returns the following day; it tends to 'regress' towards its average price, violating EMH. Lastly, after a day where CBA excess returns exceeded 2 standard deviations above the mean, excess returns are on average 0.26 percentage points higher the next day. The above results are not consistent with EMH, and all exhibit weak-form. An investor attempting to 'beat the market' through technical analysis may do so by discovering these behavioural biases, and use past information such as the CSL extreme days, BHP's cumulative log return and the cumulative returns 2 standard deviations above the mean to make inferences about future prices at time *t.*

Robustness

Following the baseline Multi-Factor CAPM model, several robustness checks were conducted to evaluate the stability of the findings. Financial return data often violates the classical OLS assumptions of homoskedasticity and independent errors due to volatility clustering and autocorrelation. We adjust for these concerns through the application of Newey-West HAC standard errors, a post-COVID subsample analysis, along with an alternative specification on the magnitude of RBA rate cuts and hikes.

Factor / Variable Name	Original OLS Model	Newey-West HAC Robust Model
BHP		
Market Excess Return (β)	1.21902*** (0.02736)	1.21902*** (0.06226)
Rate Cut	-0.00929* (0.00453)	-0.00929 (0.00930)
Momentum	-0.00851* (0.00358)	-0.00851 (0.00611)
CBA		
Market Excess Return (β)	1.10940*** (0.01591)	1.10940*** (0.02971)
Post Hike Day 1	-0.00435* (0.00219)	-0.00435 (0.00291)
Overreaction (+)	0.00257* (0.00113)	0.00257 (0.00184)
CSL		
Market Excess Return (β)	0.83820*** (0.02398)	0.83820*** (0.03762)
Overreaction (+)	-0.00424** (0.00160)	-0.00424 (0.00324)
Overreaction (-)	0.00590*** (0.00150)	0.00590 (0.00392)
WES		
Alpha (Intercept)	0.00046* (0.00019)	0.00046* (0.00018)
Market Excess Return (β)	0.89512*** (0.01893)	0.89512*** (0.03281)

Values display point estimates followed by standard errors in parentheses.
R-squared: CBA (0.6431), WES (0.4523), BHP (0.4245), CSL (0.3203)
*** p<0.001, ** p<0.01, * p<0.05

Figure 4: HAC VS Baseline Model

Robustness Test 1: We employed the Newey-West HAC as a standard error correction for autocorrelation and heteroskedasticity; these are both violations of OLS assumptions. In this correction, the only values corrected will be the p-values. Examining Figure 4, the beta values for all stocks and the WES alpha remained significant, representing the most robust finding. All other variables that previously had significance in the regression analysis are no longer significant, demonstrating that some of those results may have been driven by noise/overstated by the CAPM. Under Newey-West HAC, there is only evidence of semi-strong form EMH for one stock, suggesting conclusions should be considered with caution.

Factor / Variable Name	Full Baseline Model	Post-COVID Subperiod Model
BHP		
Market Excess Return (β)	1.21902*** (0.02736)	1.20264*** (0.05242)
Rate Cut	-0.00929* (0.00453)	0.00000 (0.00000)
Momentum	-0.00851* (0.00358)	-0.00065 (0.00616)
Overreaction (-)	-0.00043 (0.00255)	-0.00839** (0.00309)
CBA		
Alpha (Intercept)	0.00025 (0.00016)	0.00079** (0.00029)
Market Excess Return (β)	1.10940*** (0.01591)	1.03818*** (0.03385)
Post Hike Day 1	-0.00435* (0.00219)	-0.00490* (0.00240)
Overreaction (+)	0.00257* (0.00113)	0.00312 (0.00288)
CSL		
Market Excess Return (β)	0.83820*** (0.02398)	0.72455*** (0.04396)
Overreaction (+)	-0.00424** (0.00160)	0.00223 (0.00355)
Overreaction (-)	0.00590*** (0.00150)	-0.00313 (0.00283)
WES		
Alpha (Intercept)	0.00046* (0.00019)	0.00033 (0.00032)
Market Excess Return (β)	0.89512*** (0.01893)	0.97839*** (0.03837)

Values display point estimates followed by standard errors in parentheses.
R-squared: CBA (0.4868), WES (0.3923), BHP (0.3476), CSL (0.2090)
*** p<0.001, ** p<0.01, * p<0.05 based on two-tailed tests.

Figure 5: Sub-Period Results VS Baseline Model

Robustness Test 2: A post-COVID subsample analysis was conducted using data from 2021(01/01/2021) to 2024, corresponding to the period of RBA monetary policy tightening, in an attempt to better understand the effects of information in the market. This involved isolating the analysis to interest rate hikes only, reducing the total number of observations to 1011. Figure 5 illustrates which securities remained significant, and the changes. Some notable findings include a significant CBA alpha at the 1% level (p=0.00677), lost significance for WES and CSL, and a significant coefficient for large positive returns of BHP. During this period of hiking, CBA has been mispriced and is expected to have a return that is 0.0795% per day higher than the CAPM predicts, and the WES and CSL coefficients may be driven by other factors, with WES's overreaction likely driven by the volatility of the full sample period. Lastly, days that were largely negative in the COVID sample were followed by more negative returns for BHP. These results raise slight concern for the original CAPM's validity, with a change in the subsample revealing an efficient market for WES and CSL, contrary to the original.

Factor / Variable Name	Baseline Model	Rate Change Magnitude Specification
BHP		
Market Excess Return (β)	1.21902*** (0.02736)	1.22073*** (0.02736)
Rate Cut	-0.00929* (0.00453)	0.00147 (0.00940)
Momentum	-0.00851* (0.00358)	-0.00868* (0.00358)
CBA		
Market Excess Return (β)	1.10940*** (0.01591)	1.11006*** (0.01590)
Post Hike Day 1	-0.00435* (0.00219)	-0.00433* (0.00219)
Overreaction (+)	0.00257* (0.00113)	0.00249* (0.00113)
CSL		
Market Excess Return (β)	0.83820*** (0.02398)	0.83663*** (0.02398)
Overreaction (+)	-0.00424** (0.00160)	-0.00413** (0.00160)
Overreaction (-)	0.00590*** (0.00150)	0.00587*** (0.00151)
WES		
Alpha (Intercept)	0.00046* (0.00019)	0.00048* (0.00019)
Market Excess Return (β)	0.89512*** (0.01893)	0.89463*** (0.01891)

Values display point estimates followed by standard errors in parentheses.
R-squared: CBA (0.6431), WES (0.4524), BHP (0.4238), CSL (0.3195)
*** p<0.001, ** p<0.01, * p<0.05

Figure 6: Magnitude Specification VS Baseline Model

Robustness Test 3: An alternative specification was considered. The baseline model treated RBA decisions as binary variables indicating whether rates increased or decreased. To test whether markets respond to the magnitude of monetary policy changes, rather than solely direction, an alternative specification was estimated using the actual change in basis points for each RBA announcement. There was limited statistical significance across the sample, shown in Figure 6. This suggests that financial markets may adjust to the direction of policy decisions rather than the exact size of the rate adjustment itself. Importantly, the core conclusions regarding market efficiency and systematic risk remained broadly unchanged from the original model.

Discussion, limitations, conclusion

Discussion of what we can and cannot conclude from our research:

The analysis of CBA, CSL, BHP and WES from 2014-2024 demonstrates a market that is characterised by mostly, but not perfectly, efficient pricing. The most robust and consistent conclusion is that the systematic risk (Beta) is estimated to be the key driver of stock returns and remains statistically significant across all robustness checks. Additionally, our research identifies specific instances of semi-strong form market inefficiency. Most notably, WES alpha (remained robust after applying Newey-West HAC adjustments), implies that WES exhibited statistically significant abnormal returns relative to the CAPM benchmark.

Conversely, the findings also illustrate that evidence of inefficiency is heterogeneous and fragile. The OLS models identified significant behavioural relationships, including the CLS's overreaction effect and BHP's mean-reverting momentum pattern. However, several of these findings weakened or disappeared after adjusting for heteroskedasticity and autocorrelation using Newey-West HAC standard errors. Likewise, the post-COVID subsample analysis suggests that many observed relationships appear to be sensitive to the underlying macroeconomic environment. For example, CSL and WES lost significance during the 2021-2024 subsample, implying that the earlier 'abnormal' returns may have been influenced by the low interest rate and high liquidity conditions of the pre-COVID period. Consequently, the identified behaviour patterns should be interpreted as conditional correlations rather than universal or causal relationships.

Limitations:

Several limitations should be acknowledged when interpreting the findings of this analysis. First, the CAPM framework is inherently restrictive, as it explains returns primarily through systematic market risk. Important factors such as firm size, value characteristics, commodity prices, exchange rates and global macroeconomic conditions were not explicitly incorporated into the model. This creates the possibility of Omitted Variable Bias (OVB), particularly for internationally exposed firms such as CSL and commodity-sensitive firms such as BHP.

Secondly, the use of binary RBA policy variables simplifies the information contained within monetary policy announcements. Financial markets often partially anticipate interest rate decisions prior to the official announcement date, meaning prices may already reflect some of the information before the event occurs. As a result, the estimated policy coefficients may understate the true market response.

Thirdly, because the analysis relies on daily data, it is difficult to fully isolate the impact of the RBA announcements from other political or macroeconomic news occurring on the same trading day. Consequently, some estimated relationships may reflect coinciding external events rather than purely monetary policy effects.

Finally, the sample only contains four ASX-listed firms operating across different industries. While this does allow for comparisons to be drawn between sectors, it does limit the broader generalisability of the findings across the Australian equity market.

Recommendations:

Future research could strengthen the analysis through several extensions. Adopting a multi-factor asset pricing model, such as the Fama-French framework, would reduce the omitted variable bias and provide a more comprehensive benchmark for abnormal returns. Furthermore, using intraday data as an alternative to daily data would allow for greater precision in the identification of market reactions to RBA announcements and reduce contamination from coinciding macroeconomic events. Lastly, industry level clustered standard errors or a broader panel data approach could improve inference reliability by accounting for correlated shocks across firms in similar sectors.

Conclusion:

This study examined whether macroeconomic information and behavioural return patterns influence stock returns within the Australian equity market between 2014 and 2024. Utilising an extended CAPM framework, incorporating RBA monetary policy announcements, momentum, and overreaction variables, the results provide mixed evidence regarding the Efficient Market Hypothesis. Systematic market risk remained consistently significant across all firms and robustness specifications, reinforcing its importance in explaining stock returns. However, several firms exhibited delayed responses to RBA announcements and evidence of behavioural predictability, suggesting that information may not always be completely and immediately incorporated into prices. Despite these findings, the robustness analysis demonstrated that several behavioural and macroeconomic effects were sensitive to inference assumptions and sample selection. Consequently, the results are more consistent with conditional and temporary inefficiencies as opposed to persistent violations of market efficiency.

Overall, the Australian equity market appears to be broadly efficient, though stock specific and time dependent inefficiencies may emerge under particular macroeconomic and behavioural conditions.

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